Mock 13: Wrong-Answer Notes

CFA Level II · 17 wrong answers, one per page

34 / 51

Mock 13 (66.7%)

Score by topic

Topic	Score	%
Alternative Investments	1/3	33%
Derivatives	2/5	40%
Economics	2/4	50%
Portfolio Management	3/6	50%
Corporate Issuers	3/5	60%
Financial Statement Analysis	5/7	71%
Fixed Income	3/4	75%
Equity Valuation	5/6	83%
Ethical and Professional Standards	7/8	88%
Quantitative Methods	3/3	100%

Questions at a glance

Q	Topic	You	Ans
Q2	Ethics	B	A
Q13	FSA	A	B
Q14	FSA	B	C
Q19	Equity	A	B
Q25	FI	A	C
Q29	Econ	B	A
Q30	Econ	A	B
Q33	Corp	A	C
Q36	Corp	C	B
Q38	Deriv	B	A
Q39	Deriv	A	C
Q41	Deriv	C	B
Q43	Alt	C	B
Q45	Alt	B	C
Q48	PM	B	C
Q50	PM	C	A
Q51	PM	A	C

Mock 13 Q2 **Fair dealing recommended procedures: FIFO order execution (bundling allowed) AND same price/commission for all accounts in a block trade**

You **B** · Answer **A**

Ethics

VIGNETTE (RELEVANT PART, UNEDITED)

One of the first things Voss did after arriving at the firm was to meet with Owen Takeda, CFA, the firm's chief compliance officer. Takeda had been with the firm only a few months. He told Voss the previous violations had primarily related to equity research, equity trading, and client confidentiality. After reviewing the summary of violations and the firm's existing handbook, Voss and Takeda decided to submit the following recommendations regarding the firm's trade allocation procedures for block trades to the board of directors for approval:

Recommendation 1

Orders will be executed on a first-in, first-out basis, with consideration given to bundling orders for efficiency.

Recommendation 2

All accounts participating in a block trade should receive the same execution price and pay the same commission rate.

Question 2. Are the recommendations submitted to the board of directors most likely consistent with the CFA Institute Recommended Procedures for Compliance with Standard III(B): Fair Dealing?

OPTIONS

A	Yes	✓ Correct	Correct: both recommendations are recommended procedures, so the answer is Yes.
B	No with regard to Recommendation 1	✗ Your pick	Your pick: first-in, first-out with bundling IS a recommended procedure, so Recommendation 1 is consistent.
C	No with regard to Recommendation 2		Recommendation 2 (same price and commission for every account in the block) is also recommended.

Trigger "Orders will be executed on a first-in, first-out basis, with consideration given to bundling orders for efficiency."
"All accounts participating in a block trade should receive the same execution price and pay the same commission rate."

Rule next time: Both recommendations are on the III(B) list: orders executed FIFO (bundling for efficiency is fine); every account in a block trade gets the same price and commission; partial fills are shared pro rata; written allocation policy and disclosure. Violations are choosing accounts after the fill (cherry-picking) or favoring some accounts.

FORMULA / RULE

Order sequence: first-in, first-out (bundling allowed)
Block trade: same price and commission for all accounts; partial fill: pro rata
Written allocation policy, disclosed to clients

WHY THE ANSWER IS A

1. Recommendation 1: FIFO with bundling = recommended.
2. Recommendation 2: same price and commission = recommended.
3. Both consistent: **A**.

Apply to other questions: Order of processing → first-in, first-out; how a block is shared → same price + pro rata for partial fills. Violation = deciding the split after the results are known.

Mock 13 Q13 **Pension cost vs 20X1 assumptions: LOWER salary growth is the only change that helps earnings**

You **A** · Answer **B**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Montclair Retail is a retailer that compensates employees with a DB pension plan and stock options in addition to short-term benefits and a DC plan. Montclair prepares its financial statements in accordance with US GAAP.

Peter Lindholm, CFA, is an equity analyst concerned with earnings quality. He is particularly interested in whether the discretionary assumptions the company is making regarding its DB plan are aggressive or conservative. Lindholm gathers the assumptions in the table below.

Assumptions Used for Montclair DB Plan

	20X3	20X2	20X1
Expected long-term rate of return on plan assets	6.10%	6.35%	6.80%
Discount rate used to estimate pension obligation at beginning of year	4.70%	4.85%	5.20%
Estimated salary growth rate	3.50%	4.10%	4.00%
Estimated inflation rate	3.20%	2.80%	2.60%

Question 13. Compared to the assumptions Montclair used to compute its periodic pension cost in 20X1, earnings in 20X3 were most favorably affected by the change in the:

OPTIONS

A	discount rate.	✗ Your pick	Your pick: a LOWER discount rate raises the obligation and, in most cases, the interest cost, which lowers earnings.
B	estimated future salary increases.	✓ Correct	Correct: salary growth fell from 4.00% to 3.50%, which lowers the service cost and raises earnings.
C	expected long-term rate of return on plan assets.		A lower expected return on plan assets means less return is credited against the pension cost, so the cost rises and earnings fall.

Trigger “earnings in 20X3 were most favorably affected”
“Expected long-term rate of return on plan assets | 6.10% | 6.35% | 6.80%”
“Estimated salary growth rate | 3.50% | 4.10% | 4.00%”
Rule next time: Compare 20X3 with 20X1 (the column two years back, not 20X2). Salary growth 4.00% → 3.50%: lower future salaries → lower service cost → higher earnings. Discount rate down (5.20% → 4.70%) raises the obligation and interest cost; expected return on assets down (6.80% → 6.10%) raises the pension cost. Both hurt earnings.

FORMULA / RULE

> Pension cost = service cost + interest cost (discount rate × beginning obligation) − expected return on plan assets
> Higher discount rate or higher expected return or lower salary growth → lower pension cost (aggressive for earnings)

WHY THE ANSWER IS B

1. Salary growth 4.00% → 3.50%: lower service cost → favorable.
2. Discount rate lower: unfavorable; expected return lower: unfavorable.
3. Only the salary growth change helps: **B**.

Apply to other questions: Aggressive (earnings-friendly) pension assumptions = higher discount rate, higher expected return, lower salary growth; read the right column pair (20X3 vs 20X1).

Mock 13 Q14 Consistency of pension assumptions: salary growth should rise and fall with inflation

You **B** · Answer **C**

FSA

VIGNETTE (RELEVANT PART, UNEDITED)

Montclair Retail is a retailer that compensates employees with a DB pension plan and stock options in addition to short-term benefits and a DC plan. Montclair prepares its financial statements in accordance with US GAAP.

Assumptions Used for Montclair DB Plan

	20X3	20X2	20X1
Expected long-term rate of return on plan assets	6.10%	6.35%	6.80%
Discount rate used to estimate pension obligation at beginning of year	4.70%	4.85%	5.20%
Estimated salary growth rate	3.50%	4.10%	4.00%
Estimated inflation rate	3.20%	2.80%	2.60%

Question 14. Compared to the pension assumptions Montclair used in 20X2, which of the following pairs of assumptions used in 20X3 is most likely internally inconsistent?

OPTIONS

A	Expected long-term rate of return on plan assets, discount rate		Expected return and discount rate were both lowered: consistent with each other.
B	Discount rate, estimated future salary increases	✗ Your pick	Your pick: discount rate and salary growth both fell (4.85% → 4.70% and 4.10% → 3.50%); that is not an inconsistency.
C	Estimated future salary increases, inflation	✓ Correct	Correct: inflation rose but salary growth fell; salary increases are positively related to inflation.

Trigger "Estimated salary growth rate | 3.50% | 4.10% | 4.00%"
"Estimated inflation rate | 3.20% | 2.80% | 2.60%"
"Compared to the pension assumptions Montclair used in 20X2"

Rule next time: Compare 20X3 with 20X2. Inflation went UP (2.80% → 3.20%) while the salary growth estimate went DOWN (4.10% → 3.50%): these two normally move together, so the pair is inconsistent. Discount rate and expected return both fell (a consistent pair); discount rate and salary growth falling together is not a contradiction.

FORMULA / RULE

> Salary growth ≈ inflation + real growth: they should move in the same direction
> Expected return on assets and discount rate: both follow interest rates

WHY THE ANSWER IS C

1. Inflation 2.80% → 3.20% (up).
2. Salary growth 4.10% → 3.50% (down): opposite directions.
3. Inconsistent pair: **C**.

Apply to other questions: Consistency check = look for assumptions that should move together but moved apart (salary growth vs inflation).

Mock 13 Q19 **Residual income = net income − required return × BEGINNING-of-year equity**

You **A** · Answer **B**

Equity

VIGNETTE (RELEVANT PART, UNEDITED)

Exhibit 1 Selected Information from SeoHan's 2013 and 2012 Financial Statements (KRW millions, except where noted)

	Year Ended 31 December 2013	
Net income	126,000	
Dividends	50,000	
Number of common shares outstanding (actual)	30,000,000	
	KRW at 31 December 2013	**KRW at 31 December 2012**
Total stockholders' equity	812,000	748,000

Exhibit 2 Other Financial Data for SeoHan Electronics

Required return on equity	12%
Market price per share at 31 December 2013	KRW69,000

Question 19. Using the information in Exhibits 1 and 2, SeoHan's residual income (in millions) in 2013 is closest to:

OPTIONS

A	KRW28,560.	✗ Your pick	Your pick: charges the cost of equity on END-of-year equity (812,000): 28,560.
B	KRW36,240.	✓ Correct	Correct: 126,000 − 0.12 × 748,000 = KRW36,240 million.
C	KRW76,000.		Net income minus dividends (76,000): no equity charge at all.

Trigger “Total stockholders' equity | 812,000 | 748,000”
“Required return on equity | 12%”
“KRW at 31 December 2012” (the beginning equity)

Rule next time: Equity charge = required return × equity at the START of the year (31 December 2012 = 748,000). RI = 126,000 − 0.12 × 748,000 = 36,240.

FORMULA / RULE

> RI = net income − r × beginning book value of equity

WHY THE ANSWER IS B

1. Beginning equity = the 2012 column: 748,000.
2. Equity charge = 0.12 × 748,000 = 89,760.
3. RI = 126,000 − 89,760 = 36,240: **B**.

Apply to other questions: Residual income uses BEGINNING equity; check which balance sheet column is the beginning one.

Mock 13 Q25 Volatility up: any option gains value; the callable bond (short the call) falls, the putable bond (long the put) rises

You **A** · Answer **C**

FI

VIGNETTE (RELEVANT PART, UNEDITED)

Hartley & Grant is a fixed-income specialty firm that offers advisory services to investment management companies. On 1 October 20X0, Wesley Park, a senior analyst at Hartley & Grant, is reviewing three fixed-rate bonds issued by a local firm, Clearwater, Inc. The three bonds, whose characteristics are given in Exhibit 4, carry the highest credit rating.

Exhibit 4: Fixed-Rate Bonds Issued by Clearwater, Inc.

Bond	Maturity	Coupon	Type of Bond
Bond #1	1 October 20X3	4.60% annual	Option-free
Bond #2	1 October 20X3	4.60% annual	Callable at par on 1 October 20X1 and on 1 October 20X2
Bond #3	1 October 20X3	4.60% annual	Putable at par on 1 October 20X1 and on 1 October 20X2

Question 25. All else being equal, if Park assumes an interest rate volatility of 15% instead of 10%, the bond that would most likely increase in value is:

OPTIONS

A	Bond #2.	✗ Your pick	Your pick: Bond #2 is callable; higher volatility raises the call's value, which lowers the callable bond's value.
B	Bond #1.		Bond #1 is option-free: volatility does not change its value.
C	Bond #3.	✓ Correct	Correct: Bond #3 is putable; the put gains value, so the putable bond rises.

Trigger “interest rate volatility of 15% instead of 10%” (question)
“Putable at par on 1 October 20X1 and on 1 October 20X2”
“Callable at par on 1 October 20X1”

Rule next time: Higher volatility raises the value of every option. Putable bond = straight + put (investor owns the option) → value UP. Callable bond = straight − call (investor is short the option) → value DOWN. Option-free bond: unchanged.

FORMULA / RULE

Callable bond value = straight bond − call option
Putable bond value = straight bond + put option
Volatility ↑ → option value ↑ (call and put)

WHY THE ANSWER IS C

1. Volatility up: both options gain value.
2. Callable (you are short the call) falls; putable (you own the put) rises.
3. Bond #3: **C**.

Apply to other questions: Ask who OWNS the option: the investor owns the put (putable) and is short the call (callable).

Mock 13 Q29 **Neoclassical model: capital flows to capital-poor countries, capital grows faster, incomes converge**

You **B** · Answer **A**

Econ

VIGNETTE (RELEVANT PART, UNEDITED)

Dieter Kaufmann is the CEO of Rhein Builders and Finance, based in Germany. The company specializes in building low-income housing in developing countries, using new, highly specialized technology that lowers labor costs. Kaufmann is considering an investment in three neighboring African countries.

Prior to leaving the region, Kaufmann meets again with the Minister of Finance to ask for tax incentives to make his investment in that country. He makes the following justifications:

Justification 1

My investment will cause capital growth to rise more quickly and will result in higher productivity growth, causing per capita incomes to converge.

Justification 2

As a result of my successful investments, more and more foreign investors will be attracted to the country. Therefore, the country will no longer be considered capital poor.

Justification 3

My high-tech investment will likely cause the local housing industry to become more efficient; some local companies will become more innovative but some will leave the industry.

Question 29. Which of Kaufmann's justifications most likely reflects the neoclassical growth model?

OPTIONS

A	Justification 1	✓ Correct	Correct: faster capital growth and convergence is the neoclassical story.
B	Justification 2	✗ Your pick	Your pick: Justification 2 says the country will no longer be capital poor; that ends the capital inflow (returns fall), it is not the neoclassical channel.
C	Justification 3		Justification 3 (inefficient firms exit, others innovate) is an endogenous growth idea.

Trigger “My investment will cause capital growth to rise more quickly and will result in higher productivity growth, causing per capita incomes to converge.”
“the country will no longer be considered capital poor”

Rule next time: Neoclassical (open economy): capital moves from capital-rich to capital-poor countries seeking higher returns, so the poor country's capital stock grows faster, productivity grows and per capita incomes converge (Justification 1). Justification 2 is wrong: once the country is no longer capital poor, returns fall and flows slow. Justification 3 is the endogenous growth selection effect.

FORMULA / RULE

> Neoclassical: capital deepening from foreign capital → convergence
> Endogenous growth: innovation, selection effect

WHY THE ANSWER IS A

1. Capital flows to where capital is scarce.
2. Capital stock rises faster → productivity growth → convergence.
3. Justification 1: **A**.

Apply to other questions: Neoclassical keywords: capital flows, capital poor, convergence; endogenous keywords: innovation, R&D, firms exit.

Mock 13 Q30 **Flow supply/demand channel: persistent current account SURPLUS → currency appreciates**

You **A** · Answer **B**

Econ

VIGNETTE (RELEVANT PART, UNEDITED)

Ananya Rao is an analyst who works in the finance department of Silverline Entertainment Group (SEG), a global entertainment company based in London, United Kingdom. SEG creates digital and film media content as well as maintaining a number of theme parks around the world.

Rao notes that Country A is currently experiencing positive economic growth, low unemployment, a high level of debt relative to GDP, and consistently maintains a current account surplus due to the demand for its commodities. Based on these factors and according to the flow supply/demand channel, Rao asserts in her analysis that Country A's currency should appreciate over time and that SEG should consider a longer-term hedging strategy.

Question 30. Rao's assertion about the change in the value of Country A's currency is most likely:

OPTIONS

A	correct because countries with trade surplus will finance their trade with increased borrowing.	✗ Your pick	Your pick: that is the portfolio balance channel, and it applies to trade DEFICITS, not surpluses.
B	correct because countries with persistent current account surpluses tend to appreciate over time.	✓ Correct	Correct: persistent current account surpluses lead to appreciation (flow supply/demand).
C	incorrect because countries have an upper limit on the ability to borrow during trade surpluses.		The upper limit on borrowing is the debt sustainability channel and applies to DEFICIT countries.

Trigger “consistently maintains a current account surplus”
“according to the flow supply/demand channel”
“currency should appreciate over time”

Rule next time: Flow supply/demand: surplus nations see more demand for their currency, so it appreciates (deficit nations depreciate). Portfolio balance: DEFICIT countries finance with borrowing, shifting asset preferences. Debt sustainability: an upper limit on persistent large DEFICITS, which forces a depreciation.

FORMULA / RULE

> Flow: surplus → appreciate, deficit → depreciate
> Portfolio balance: deficits financed by borrowing → asset preference shifts
> Debt sustainability: limit on persistent deficits → depreciation

WHY THE ANSWER IS B

1. The vignette names the flow channel and a surplus.
2. Surplus → more demand for the currency → appreciation.
3. Correct reason: **B**.

Apply to other questions: Match the channel name in the text to its rule; the other two channels talk about DEFICITS.

Mock 13 Q33 **Short-term rate in an inverted curve → both ERP and risk-free rate too high → upward bias**

You **A** · Answer **C**

Corp

VIGNETTE (RELEVANT PART, UNEDITED)

An equity index is established in Year 1 for a country that has recently moved to a market economy. The index vendor constructed returns for the four years prior to Year 1 based on the initial group of companies constituting the index in Year 1. From Year 12 to Year 16, a series of military confrontations concerning a disputed border disrupted the economy and financial markets. The dispute is conclusively arbitrated at the end of Year 16. In total, 20 years of equity market return history is available. Other selected data are in the following table.

Selected Data

Geometric mean return relative to 10-year government bond returns (over a 20-year period)	1.7% per year
Arithmetic mean return relative to 10-year government bond returns (over a 20-year period)	2.0% per year
Index forward dividend yield	1.8%
Forecasted public company earnings growth	4% per year
Forecasted market P/E growth	0.4% per year
Forecasted real GDP growth rate (by Year 19)	2.8%
Current vs. long-term inflation forecast	5.0% vs. 3.2% per year
Current yield curve (inversion)	Short maturities: 7.5%
	10-year maturities: 6.0%

Question 33. In the current interest rate environment, using a required return on equity estimate based on the short-term government bond rate and a historical ERP defined in terms of a short-term government bond rate would be expected to:

OPTIONS

A	bias long-term required return on equity estimates downward.	✗ Your pick	Your pick: both effects (larger ERP, higher risk-free rate) raise the estimate, not lower it.
B	have no effect on long-term required return on equity estimates.		The choice of risk-free rate matters, so the effect is not zero.
C	bias long-term required return on equity estimates upward.	✓ Correct	Correct: upward bias for long-term required return estimates.

Trigger “Current yield curve (inversion) | Short maturities: 7.5%”
“10-year maturities: 6.0%”
“historical ERP defined in terms of a short-term government bond rate”

Rule next time: A historical ERP measured against a short-term rate is larger (short rates are normally below long rates), and today the short rate (7.5%) is above the long rate (6.0%). Required return = risk-free rate + beta × ERP: both parts are too high → long-term estimates are biased UPWARD.

FORMULA / RULE

> Required return on equity = risk-free rate + beta × equity risk premium
> ERP against a short rate > ERP against a long rate (normal curve)

WHY THE ANSWER IS C

1. Inverted curve: short rate 7.5% > long rate 6.0%.
2. ERP measured against the short rate is also larger.
3. Both raise the estimate: **C**.

Apply to other questions: Match the ERP to the risk-free rate used: short-term ERP with a short-term rate overstates long-term required returns.

Mock 13 Q36 **Conglomerate discount = sum-of-the-parts value − trading value; a negative number is a premium**

You **C** · Answer **B**

Corp

VIGNETTE (RELEVANT PART, UNEDITED)

Cedarbrook Energy Corporation is a publicly traded, US-based energy company that just announced its acquisition of Tidewater Energy Corporation, a smaller US-based energy company. Cedarbrook's current trading enterprise value just prior to the announcement was USD146 billion.

Hartmann conducted a sum-of-the-parts valuation of Cedarbrook's three segments and calculated an estimated enterprise value of USD142 billion just prior to the announcement.

Question 36. Cedarbrook's estimated conglomerate discount just prior to the announcement is:

OPTIONS

A	USD0.		Zero would need the two values to be equal; they are 142 and 146.
B	–USD4 billion.	✓ Correct	Correct: 142 − 146 = −USD4 billion.
C	USD4 billion.	✗ Your pick	Your pick: +4 would mean the market values the company below its parts; here it is above them.

Trigger “current trading enterprise value just prior to the announcement was USD146 billion” “estimated enterprise value of USD142 billion”

Rule next time: Conglomerate discount = SOTP enterprise value − current trading enterprise value = 142 − 146 = −USD4 billion. Negative means the market values the company ABOVE the sum of its parts (a premium).

FORMULA / RULE

> Conglomerate discount = SOTP value − trading value (negative = premium)

WHY THE ANSWER IS B

1. SOTP = 142; trading = 146.
2. 142 − 146 = −4.
3. Negative sign kept: **B**.

Apply to other questions: Do the subtraction in the order SOTP − market and keep the sign.

Mock 13 Q38 **Two-period binomial call value: weight the payoffs with risk-neutral probabilities and discount TWO periods**

You **B** · Answer **A**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Lang describes the two-period binomial model to Kessler and provides information related to Birchwood Corporation's common stock and options on that stock (presented in Exhibit 1). Birchwood stock does not pay a dividend.

Exhibit 1 Information on Birchwood Corporation Stock and Options

Stock price	$75
Call strike	$75
Risk-free interest rate	4.00%
Up factor	1.20
Down factor	0.90

Question 38. The value of a two-year European call option on Birchwood Corporation shares is likely closest to:

OPTIONS

A	$9.41.	✓ Correct	Correct: 9.41.
B	$9.78.	✗ Your pick	Your pick: you discounted the expected payoff only once (÷ 1.04); a two-year option is discounted twice (÷ 1.04²).
C	$10.40.		Probabilities of 0.5 instead of the risk-neutral 0.4667.

Trigger "two-year European call option" (question)
"Up factor | 1.20", "Down factor | 0.90"
"Risk-free interest rate | 4.00%"
Rule next time: π = (1 + r − d) ÷ (u − d) = (1.04 − 0.90) ÷ 0.30 = 0.4667. Stock at year 2: 108, 81, 60.75 → call payoffs 33, 6, 0. Value = [π² × 33 + 2π(1 − π) × 6] ÷ 1.04² = 9.41.

FORMULA / RULE

$\pi = (1 + r - d) \div (u - d)$

$C0 = [\pi^2 C_{uu} + 2\pi(1 - \pi) C_{ud} + (1 - \pi)^2 C_{dd}] \div (1 + r)^2$

WHY THE ANSWER IS A

1. π = 0.4667.
2. Expected payoff = 0.2178 × 33 + 0.4978 × 6 = 10.17.
3. ÷ 1.0816 = 9.41: **A**.

Apply to other questions: Two periods → discount by (1 + r)² (or work back node by node); never use 0.5 for the probabilities.

Mock 13 Q39 **Compare the model value with the market price: price below value → buy the call and short the replicating portfolio**

You **A** · Answer **C**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Lang describes the two-period binomial model to Kessler and provides information related to Birchwood Corporation's common stock and options on that stock (presented in Exhibit 1). Birchwood stock does not pay a dividend.

Exhibit 1 Information on Birchwood Corporation Stock and Options

Stock price	$75
Call strike	$75
Risk-free interest rate	4.00%
Up factor	1.20
Down factor	0.90

The price of a one-year call option on Birchwood Corporation common stock is $5.60. Based on this price and on information provided in Exhibit 1, Pryor, Kessler, and Lang believe that an arbitrage opportunity may exist, which arbitrageurs could exploit based on the binomial model no-arbitrage approach. Pryor states the arbitrageurs should buy call options using borrowed funds. Kessler disagrees, stating that arbitrageurs should short-sell shares and lend the proceeds. Lang also disagrees, stating that arbitrageurs should buy shares using borrowed funds.

Question 39. With respect to the one-year call option on Birchwood stock, whose strategy is most likely correct?

OPTIONS

A	Lang's	✗ Your pick	Your pick: buying shares with borrowed funds replicates a long call; it fits an OVERPRICED call (which you would sell).
B	Pryor's		Buying calls with borrowed funds is the right side on the call but the hedge needs short shares with the proceeds lent.
C	Kessler's	✓ Correct	Correct: buy the call, short-sell shares and lend the proceeds.

Trigger “The price of a one-year call option on Birchwood Corporation common stock is $5.60”
“Stock price | $75”, “Call strike | $75”

Rule next time: One-year model value = π × (90 − 75) ÷ 1.04 = $6.73 > market price $5.60: the call is underpriced. Buy the call and sell the replicating portfolio: short-sell shares and lend the proceeds (Kessler). Buying shares with borrowed funds replicates a LONG call, so it is used when the call is overpriced.

FORMULA / RULE

> Long call = buy shares + borrow; short the replicating portfolio = short shares + lend
> Market price < model value → underpriced → buy

WHY THE ANSWER IS C

1. Model value 6.73 vs price 5.60.
2. Underpriced: buy the call.
3. Hedge by shorting shares and lending: Kessler, **C**.

Apply to other questions: Direction rule: underpriced → buy it and short its replication; overpriced → sell it and buy its replication.

Mock 13 Q41 **Cost of carry for bond futures: a LATER coupon gives a higher futures price; futures below the carry price → buy futures**

You **C** · Answer **B**

Deriv

VIGNETTE (RELEVANT PART, UNEDITED)

Ingrid Hale is a portfolio manager at Lakecrest Capital, a hedge fund that frequently uses derivatives to hedge or speculate. She works with Lars Nilsen, a junior analyst.

Carry Arbitrage Model

Hale and Nilsen discuss a futures contract in which the underlying bond is expected to make an interest payment in three months.

Statement 1: Based on the cost of carry model, the futures price would be higher if the underlying bond's upcoming interest payment was expected in six months instead of three.

Statement 2: If the futures price is less than the price suggested by the carry arbitrage model, the futures contract should be purchased.

Question 41. Which of Hale's statements is correct?

OPTIONS

A	Only Statement 1		Statement 2 is also correct.
B	Both Statement 1 and Statement 2	✓ Correct	Correct: both statements are correct.
C	Only Statement 2	✗ Your pick	Your pick: Statement 1 is also correct (later coupon, smaller PV of coupons, higher futures price).

Trigger “the futures price would be higher if the underlying bond's upcoming interest payment was expected in six months instead of three”
“If the futures price is less than the price suggested by the carry arbitrage model, the futures contract should be purchased.”
Rule next time: F0 = FV[B0 + AI0 − PV(coupons)]. A coupon paid later has a smaller present value, so PVCI is smaller and the futures price is HIGHER: Statement 1 is correct. A futures price below the model price is underpriced: buy the futures (reverse carry arbitrage): Statement 2 is correct.

FORMULA / RULE

> F0 = FV[B0 + AI0 − PVCI]
> Futures < carry price: buy futures; futures > carry price: sell futures

WHY THE ANSWER IS B

1. Coupon later → PV of coupons smaller → F0 larger: Statement 1 true.
2. Underpriced futures → buy: Statement 2 true.
3. Both: **B**.

Apply to other questions: Test each statement separately with the formula before choosing 'only one'.

Mock 13 Q43 **Hedge fund selection: step 1 variance screen (SD limit), step 2 Sortino ratio for large negative events**

You **C** · Answer **B**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Rosa Almeida is an investment analyst for the Atlantic Teachers' Retirement Fund (ATRF). ATRF's investment committee (IC) has asked Almeida to evaluate a hedge fund allocation. Almeida simulates adding a 20% allocation to individual hedge fund strategies. The IC requires (1) that the variance of the combined portfolio be less than 85% of that of the current portfolio and (2) that the combined portfolio maximize the risk-adjusted return with the expectation of large negative events. Exhibit 1 provides the simulated results.

Exhibit 1: Simulated Combined Portfolios

Hedge fund strategy	Standard deviation (%)	Sharpe ratio	Sortino ratio	Maximum drawdown (%)
Current portfolio				
NA	9.20	0.46	1.00	18.10
Three potential portfolios with a 20% hedge fund allocation				
Merger arbitrage	8.30	0.60	1.66	8.20
Systematic futures	8.75	0.72	1.81	10.40
Equity market neutral	8.40	0.65	1.57	12.10

Question 43. Based on the IC's two requirements, which simulated portfolio in Exhibit 1 is most suitable?

OPTIONS

A	Systematic futures		Highest Sortino (1.81) but SD 8.75% is above the 8.48% limit.
B	Merger arbitrage	✓ Correct	Correct: passes the variance screen and has the higher Sortino of the two survivors.
C	Equity market neutral	✗ Your pick	Your pick: equity market neutral passes the screen, but its Sortino (1.57) is lower than merger arbitrage (1.66).

Trigger "variance of the combined portfolio be less than 85% of that of the current portfolio" "expectation of large negative events"

Rule next time: Variance limit: 0.85 × 9.20² = 71.94, so SD ≤ 8.48%. Systematic futures (8.75%) fails; merger arbitrage (8.30%) and equity market neutral (8.40%) pass. Large negative events expected → compare Sortino ratios (downside risk): merger arbitrage 1.66 beats equity market neutral 1.57.

FORMULA / RULE

> SD limit = √(variance fraction) × current SD
> Large negative events → Sortino ratio (downside deviation), not Sharpe

WHY THE ANSWER IS B

1. Limit SD = √0.85 × 9.20 = 8.48%.
2. Survivors: merger arbitrage, equity market neutral.
3. Higher Sortino: merger arbitrage, **B**.

Apply to other questions: Always screen on the stated constraint first, then rank the survivors with the measure the question points to.

Mock 13 Q45 Hedge fund selection: variance screen first (here 90%), then Sortino ratio

You **B** · Answer **C**

Alt

VIGNETTE (RELEVANT PART, UNEDITED)

Ellis Marwood is a portfolio analyst for the Harbor-Ridge Pension Fund (HRPF). HRPF's investment committee (IC) asks Marwood to research adding hedge funds to the HRPF portfolio.

Exhibit 2: Performance of Various Combined Portfolios

Hedge fund strategy	Standard deviation (%)	Sharpe ratio	Sortino ratio	Maximum drawdown (%)

Question 45. Based on the IC's primary considerations for a combined portfolio, which simulated hedge fund strategy portfolio in Exhibit 2 creates the most suitable combined portfolio?

OPTIONS

A	Merger arbitrage		Highest Sortino (1.88) but SD 8.35% is above the 8.16% limit.
B	Systematic futures	✗ Your pick	Your pick: systematic futures passes the screen, but its Sortino (1.74) is lower than equity market neutral (1.86).
C	Equity market neutral	✓ Correct	Correct: passes the screen and has the higher Sortino ratio.

Trigger “variance of the combined portfolio must be less than 90% of that of the current portfolio” “expectation of large negative events”

Rule next time: Variance limit: $0.90 \times 8.60^2 = 66.56$, so SD ≤ 8.16%. Merger arbitrage (8.35%) fails despite the highest Sortino (1.88). Systematic futures (7.30%) and equity market neutral (7.65%) pass; Sortino 1.86 vs 1.74: equity market neutral.

FORMULA / RULE

SD limit = $\sqrt{0.90} \times 8.60 = 8.16\%$

Sortino = excess return ÷ downside deviation

WHY THE ANSWER IS C

1. Screen: SD ≤ 8.16%.
2. Survivors: systematic futures 7.30, equity market neutral 7.65.
3. Higher Sortino: equity market neutral 1.86: **C**.

Apply to other questions: Same two steps as Q43; compute the SD limit before looking at the ratios.

Mock 13 Q48 Skill in achieving active returns = highest information ratio, not highest information coefficient

You **B** · Answer **C**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Gareth Armitage is chief investment officer at Crown & Vale Capital. Armitage decides to add a fourth offering to his group of balanced funds, Fund W, which will use a custom benchmark of 60% global equities and 40% global bonds.

Armitage conducts a search to hire a manager for the global equity portion of Fund W and identifies three candidates. He asks the candidates to prepare risk and return forecasts relative to Fund W's benchmark based on their investment strategy, with the only constraint being no short selling. Each candidate develops independent annual forecasts with active return projections that are uncorrelated and constructs a portfolio made up of stocks that are diverse both geographically and across economic sectors. Selected data for the three candidates' portfolios are presented in Exhibit 1.

Exhibit 1: Forecasted Portfolio Data for Equity Portion of Fund W

	Candidate A	Candidate B	Candidate C
Rebalancing	Annually	Annually	Annually
Number of securities	100	64	36
Information ratio (IR)	0.738	0.651	0.630
Transfer coefficient (TC)	0.820	0.740	0.500
Information coefficient*	0.09	0.11	0.21

Question 48. Based on the data presented in Exhibit 1, the candidate with the greatest skill at achieving active returns appears to be:

OPTIONS

A	Candidate B.		Candidate B: higher IC than A (0.11) but lower IR (0.651).
B	Candidate C.	✗ Your pick	Your pick: Candidate C has the highest IC (0.21) but the lowest IR (0.630): low TC and small breadth offset it.
C	Candidate A.	✓ Correct	Correct: Candidate A has the highest IR.

Trigger “greatest skill at achieving active returns” (question)
“Information ratio (IR) | 0.738 | 0.651 | 0.630”

Rule next time: The information ratio (active return per unit of active risk, IR = TC × IC × √breadth) measures the skill of generating active returns. Candidate A has the highest IR (0.738). Candidate C has the highest IC (0.21) but a low TC (0.50) and small breadth.

FORMULA / RULE

> IR = TC × IC × √BR
> Higher IR = more skill (risk-adjusted active return)

WHY THE ANSWER IS C

1. IR: A 0.738, B 0.651, C 0.630.
2. IC alone ignores TC and breadth.
3. Highest IR: Candidate A, **C**.

Apply to other questions: 'Skill' / 'consistency of active return' → information ratio; do not pick the highest IC.

Mock 13 Q50 **Recession: central banks cut short rates, long rates fall less → a flat curve becomes upward sloping**

You **C** · Answer **A**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Julia Whitmore is a financial planner at a large wealth management firm. One of her clients, Tobias Hartwell, just received a sizable inheritance. He invests a portion of the inheritance in an annuity that will immediately increase his income by a substantial amount. He enlists Whitmore's help to invest the remaining amount of the inheritance.

Exhibit 1: Selected Macroeconomic Data

	Nominal GDP Growth	Inflation Rate	Volatility of Real GDP Growth	Yield Curve Shape	Trailing 12-Month Equity Index P/E
Country #1	7.0%	4.5%	High	Flat	16.5
Country #2	5.5%	3.0%	Low	Upward slope	17.3
Country #3	4.0%	2.5%	Low	Flat	18.2

In her analysis, Whitmore observes that the spread between the three-year default-free nominal bond and the default-free real zero-coupon bond in Country #3 is 2.5%.

Whitmore predicts that Country #3 will slip into a recession next quarter. She thinks it will be short-lived, lasting only 12 months or so, and considers the impact of such a recession on the performance of the country's stocks and bonds.

Question 50. If Whitmore's prediction about the economy of Country #3 is realized, the yield curve in Country #3 will most likely:

OPTIONS

A	become upward sloping.	✓ Correct	Correct: upward sloping.
B	remain flat.		A flat curve would need short and long rates to fall by the same amount.
C	become downward sloping.	✗ Your pick	Your pick: a downward sloping curve is typical of the LATE expansion, when the central bank raises rates.

Trigger "Country #3 | 4.0% | 2.5% | Low | Flat"
"Whitmore predicts that Country #3 will slip into a recession next quarter."
"short-lived, lasting only 12 months or so"

Rule next time: The curve is flat now. In a recession the central bank cuts policy rates, so short-term yields fall a lot; long-term yields fall less because short rates are expected to return to normal as the recession ends. Short rates fall more than long rates → the curve becomes UPWARD sloping.

FORMULA / RULE

> Rates cut (recession): curve steepens (upward sloping)
> Rates raised (late expansion): curve flattens, then inverts

WHY THE ANSWER IS A

1. Recession → policy rate cut.
2. Short yields fall by more than long yields.
3. Curve slopes upward: **A**.

Apply to other questions: Use your table: recession = rates cut = upward sloping; late expansion = rates raised = flat or inverted.

Mock 13 Q51 Yields up → shorten duration, as far as the 40%-60% bucket limits allow

You **A** · Answer **C**

PM

VIGNETTE (RELEVANT PART, UNEDITED)

Pierre Delacroix is the portfolio manager of the proprietary fixed-income portfolio of Coastal Regional Bank. The portfolio is structured as a barbell portfolio: About half of the portfolio is invested in zero-coupon Treasuries with maturities in the 3- to 5-year range (Portfolio P1), and the remainder is invested in zero-coupon Treasuries with maturities in the 10- to 15-year range (Portfolio P2). Delacroix has discretion to allocate between 40% and 60% of the assets to each maturity "bucket." He must remain fully invested at all times. Exhibit 2 shows details of this portfolio.

Exhibit 2: US Treasury Barbell Portfolio

	P1 (3–5 Years)	P2 (10–15 Years)
Average duration	3.30	11.07
Market value	$46.2 million	$63.8 million

Question 51. If Delacroix is expecting a 50 bp increase in yields at all points along the yield curve, which of the following trades is he most likely to execute to minimize his risk?

OPTIONS

A	Sell $30 million of P2 and reinvest the proceeds in three-year bonds	✗ Your pick	Your pick: selling $30 million leaves 33.8 ÷ 110.0 = 30.7% in P2, below the 40% minimum.
B	Sell $8 million of P2 and reinvest the proceeds in three-year bonds		Selling only $8 million stays within limits but shortens duration far less than $19 million.
C	Sell $19 million of P2 and reinvest the proceeds in three-year bonds	✓ Correct	Correct: $19 million leaves P2 at 40.7%, just above the 40% limit, and cuts duration the most.

Trigger “Delacroix has discretion to allocate between 40% and 60% of the assets to each maturity "bucket.""
“50 bp increase in yields”
“Market value | $46.2 million | $63.8 million”

Rule next time: Total = 46.2 + 63.8 = $110.0 million. Yields rising → shorten duration: sell the long bucket (P2, duration 11.07) and buy 3-year bonds. P2 must stay at 40% or more of the total: 44.0 million would be the floor, so at most about $19.8 million can be sold. Selling $19 million leaves 44.8 ÷ 110.0 = 40.7%.

FORMULA / RULE

> Duration down when yields rise; keep each bucket between 40% and 60% of total market value

WHY THE ANSWER IS C

1. Total $110.0 million; P2 floor = 40% × 110 = 44.0.
2. Maximum sale ≈ 63.8 − 44.0 = 19.8.
3. $19 million: **C**.

Apply to other questions: Optimize inside the constraint: use the largest trade that keeps every bucket within its limit.